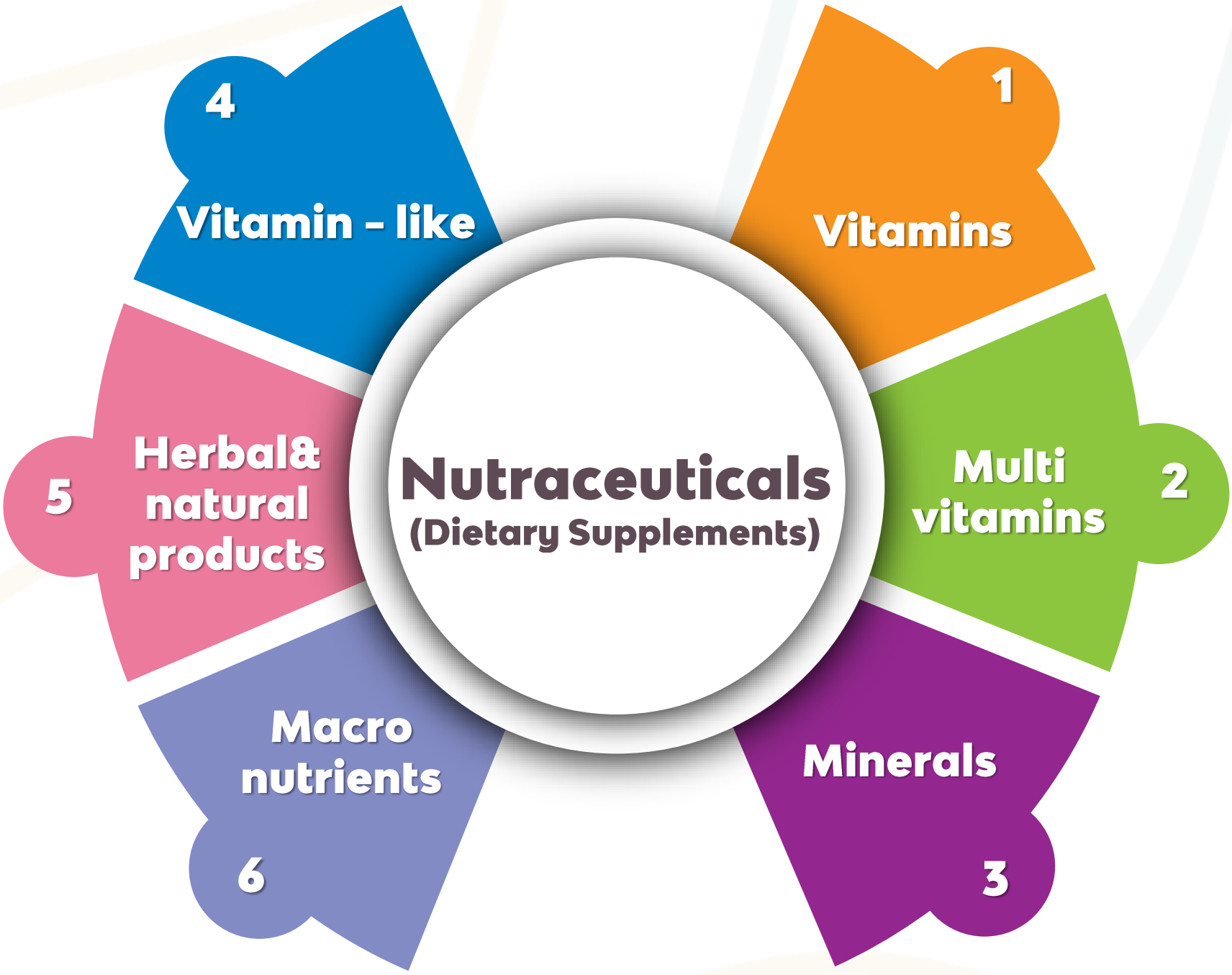
Nutraceuticals
(Dietary Supplements)



Talent Management Department

صيدليـة المتحـدة
United Pharmacy

Nutraceuticals drugs classification



Today we will learn about

01 Vitamins

02 Multivitamins

03 Minerals

04 Herbal & natural products

05 Vitamin-like

06 Macronutrients

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Nutrients & Dietary Supplements

Nutrients are the essential components found in food that your body needs to function, grow, and stay healthy. They can be broadly categorized into the following types:

- **Macronutrients (needed in larger amounts):**
  - Carbohydrates
  - Proteins
  - Fats
- **Micronutrients (needed in smaller amounts):**
  - Vitamins
  - Major Minerals
  - Trace elements

Nutritional disorders:

Nutritional disorders are health conditions caused by an imbalance or deficiency in nutrients needed by the body. Some common types of nutritional disorders include:

- Deficiency Disorders
- Excessive Intake Disorders
- Metabolic Disorders
- Genetic Disorders

Dietary Supplements in Community Pharmacy Practice

Dietary supplements in community pharmacy practice is indicated to prevents or treats nutritional disorders and other health benefits like prevention of chronic disease complications.

They could be categorized into:

- Vitamins
- Minerals
- Multivitamins
- Herbal & natural products
- Vitamin-like
- Macronutrients

Vitamins & Multivitamins

- Vitamins are micronutrients essential for various bodily functions, including growth, maintenance, and reproduction
- They are classified into two broad categories based on their solubility:

1. **Fat-soluble Vitamins:**
    - These include **Vitamins A, D, E, and K**
    - **Fat-soluble vitamins are stored in body tissues**, which means that ingesting excessive quantities can lead to toxicity

2. **Water-soluble Vitamins:**
    - These include Vitamin C (ascorbic acid) and the B-complex vitamins (riboflavin, thiamin, B6, B12, niacin, pantothenic acid, biotin, and folic acid
    - These vitamins are generally **not stored in the body**, and excessive quantities are typically excreted in the urine, making daily intake desirable for optimal health

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Vitamins & Multivitamin

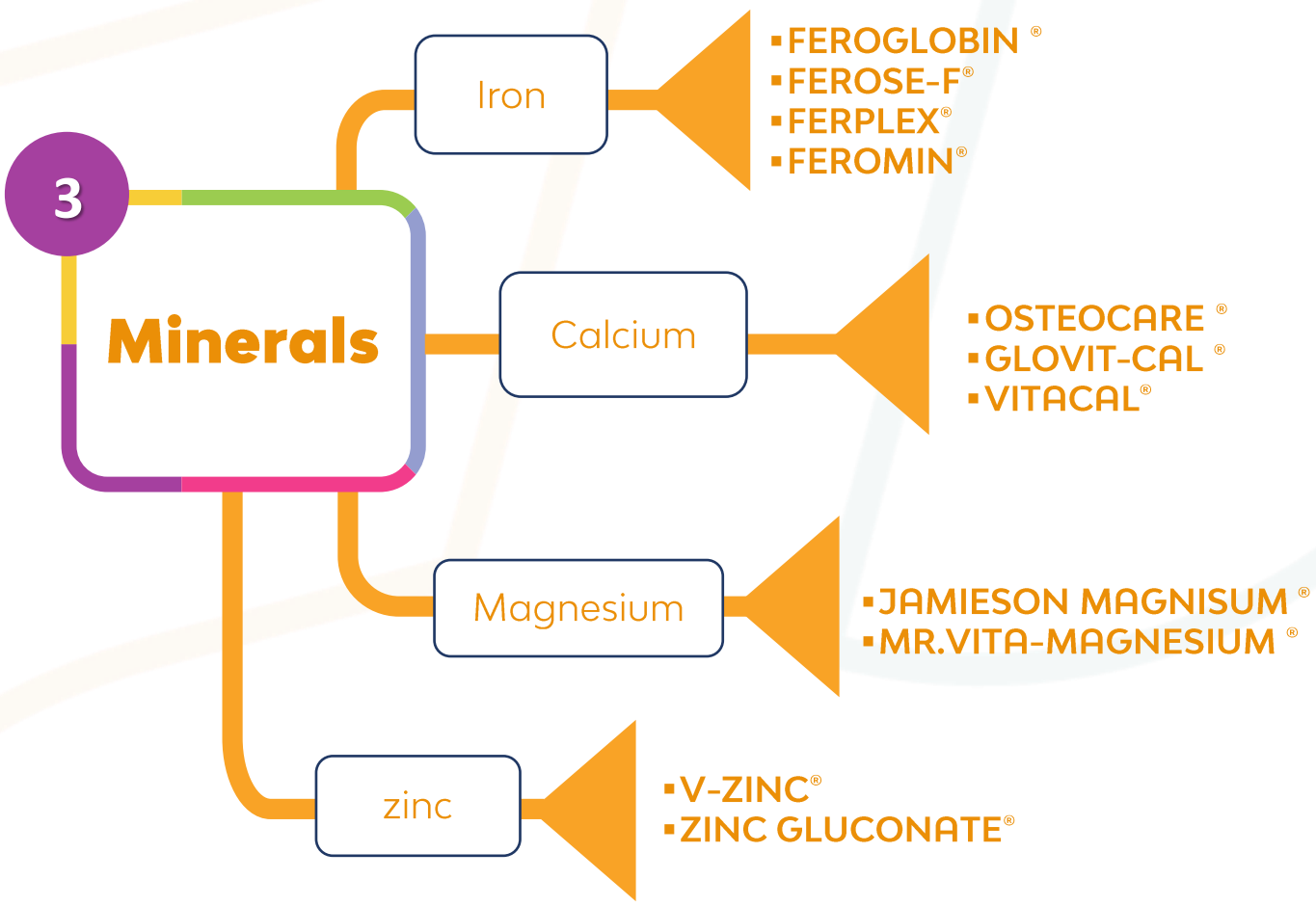
- Multivitamins
  - GLOVIT SOFTGEL®
  - CENTRUM®
  - WELLMAN®
  - PHARMATON®
  - VITAKID MULTIVITAMIN®
- B complex
  - NEUROBION®
  - METHYCOBAL.®
  - NEUROVIT®
  - NEURO-B®
- Vitamin C
  - REDOXON SUPPLEMENT®
  - EFFERVIT-C®
  - MR. VITAMIN C®
- Folic Acid
  - FOLIC ACID®
  - FOLICUM®
  - RAFOL®
- Vitamin E
  - NATURAL VIT-E®
  - EVIT®
  - VITAL-E®
- VITAMIN D
  - VIDROP VITAMIN D3®
  - PRIMA D3 50000®
  - JP VITAMIN D3®
  - D-CURA®
- VITAMIN K2
  - PRIMA K2®

Minerals

Minerals are essential **micronutrients** that constitute approximately **4% of body weigh**

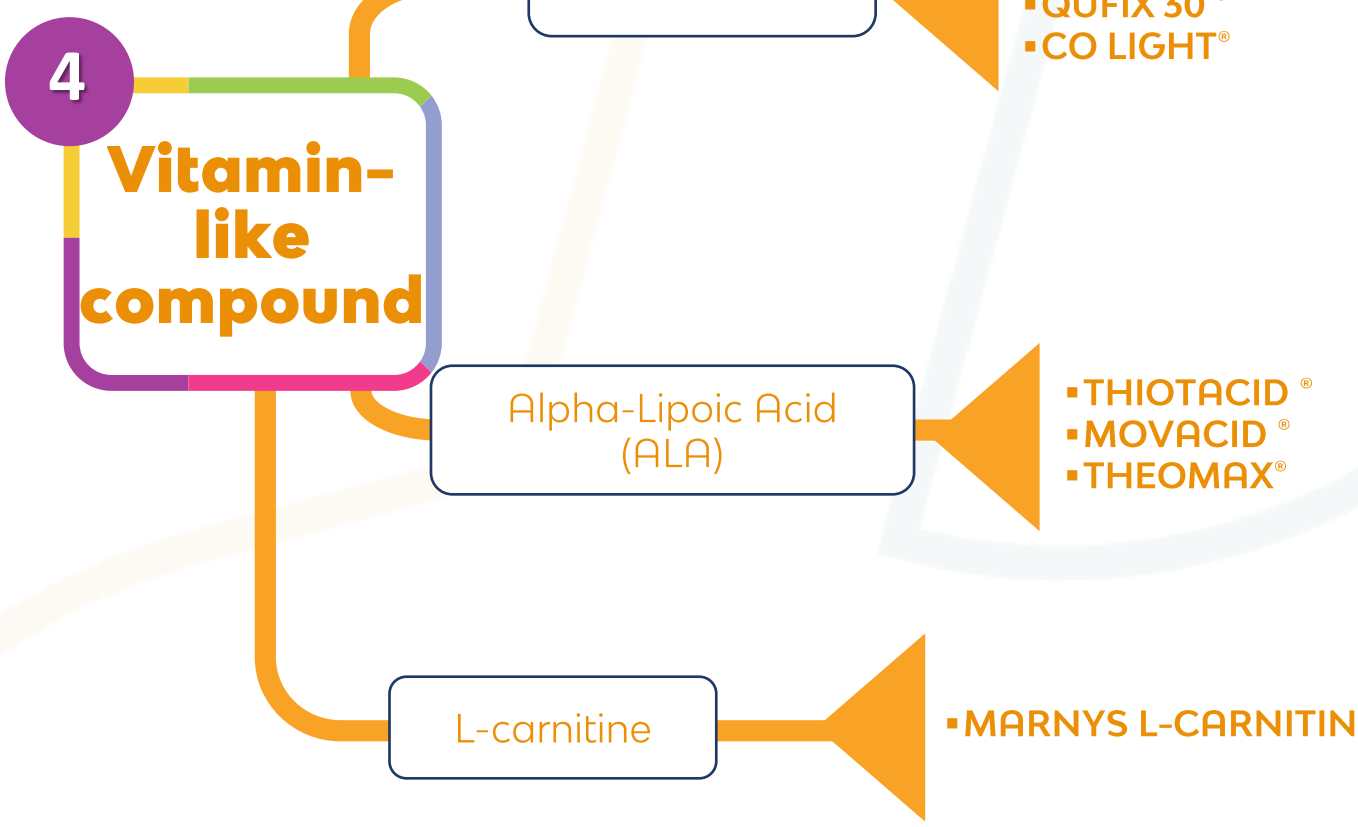
Functions and Role in the Body:

- Minerals serve diverse and crucial functions in the body:
    1. They act as **constituents of many enzymes, hormones, ... a**s well as inorganic compounds like sodium chloride, potassium chloride, calcium, and phosphorus, which are present as free ions
    2. They are involved in **regulating cell membrane permeability, osmotic pressure, and acid-base and water balance** ertain mineral ions mediate **action potential conduction and neurotransmitter action**
- Different body tissues contain varying quantities of different minerals; for instance, bone is rich in calcium, phosphorus, and magnesium, while soft tissue has a high concentration of potassium


3
Minerals
Iron
FEROGLOBIN ®
FEROSE-F®
FERPLEX®
FEROMIN®
Calcium
OSTEOCARE ®
GLOVIT-CAL ®
VITACAL®
Magnesium
JAMIESON MAGNISUM ®
MR.VITA-MAGNESIUM ®
zinc
V-ZINC®
ZINC GLUCONATE®

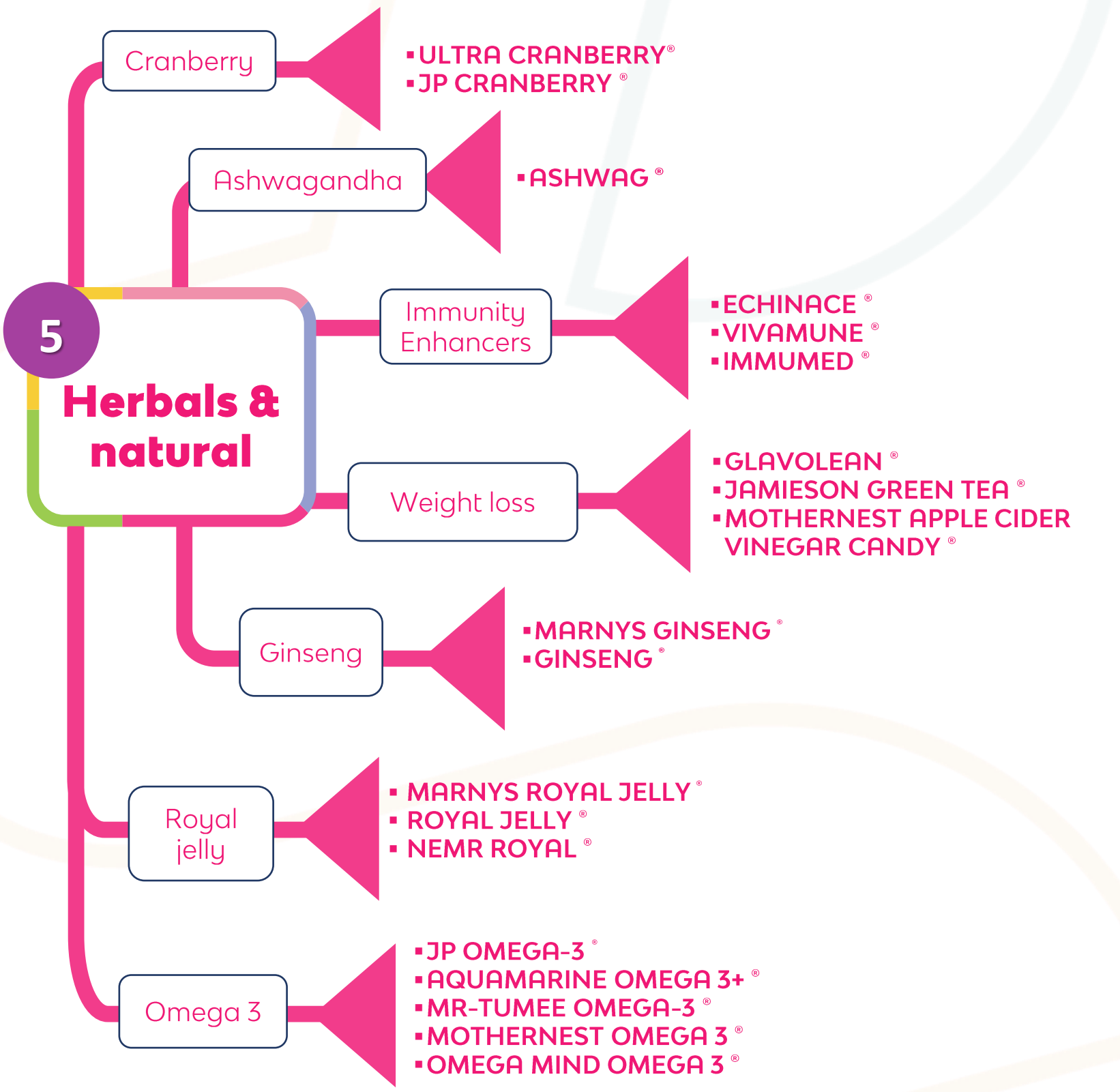
Vitamin-like

- Vitamin-like compounds are naturally occurring substances that support critical metabolic, cellular, and enzymatic processes in the human body. Although the body can **synthesize them in small amounts**, their production may be **insufficient under certain physiological or pathological conditions**, making **dietary intake or supplementation beneficial or necessary**.
- **Vitamin-like compounds** share many functional similarities with vitamins, such as:
  - Acting as **coenzymes or cofactors** in vital biochemical reactions
  - Supporting **antioxidant defense**, **energy metabolism**, and **cellular repair**
  - Becoming **conditionally essential**, especially during illness, stress, aging, or metabolic disorders
- However, they are **not classified as essential vitamins** because:
  - They can be **synthesized endogenously** (to some extent)
  - Their **deficiency does not always result in classic vitamin deficiency diseases**
  - **E.g.** Coenzyme Q10 (CoQ10), L-Carnitine and Alpha-Lipoic Acid (ALA)


4
Vitamin-like compound
Coenzyme Q10
CO-Q10®
NUTRAQUINOL®
QUFIX 30 ®
CO LIGHT®
Alpha-Lipoic Acid (ALA)
THIOTACID ®
MOVACID ®
THEOMAX®
L-carnitine
MARNYS L-CARNITINE®

Herbal dietary supplements

Herbal supplements are **concentrated preparations** of botanical materials intended to **complement the diet** and **promote specific health outcomes**, such as boosting immunity, improving digestion, reducing stress, supporting hormonal balance or for weight loss.



Macronutrients

Macronutrients are essential dietary components that play pivotal roles in providing energy and supporting various bodily functions. Unlike micronutrients (vitamins and minerals), which are needed in smaller quantities, macronutrients are required in larger amounts.

The three main types of macronutrients are:

- Carbohydrates
- Fats
- Proteins

General Functions and Storage

Macronutrients are consumed for sustenance to maintain metabolism and physical activity.

When the body is not exercising, macronutrient intake is stored for later use.

Carbohydrates are first converted to glycogen and stored in muscles and the liver for rapid, immediate energy. Any remaining carbohydrates are then converted to fatty acids for storage once glycogen stores are full.

Fats are the most calorically dense component in the diet, providing 9 kcal/g, compared to 4 kcal/g for both protein and carbohydrates.

the major part of these dietary supplements is directed to athletes that to be used during the bulking or other phases (**not available in UPC**)

Some classifications considered omega-3 as a macronutrient

الآن حان دور الأنشطة
اضغط على الزر للبدء



ابدأ النشاط